

Predicting Property sales price

The data set contains 4 continuous (Lot area, Garage area, number of full bath, and number of fireplaces), 2 ordinal (Overall quality and overall condition) and 2 categorical variables (sales condition, year the property was sold)

I created a new categorical variable called YearBuilt_cat, to capture the effect of age of property.

1872-1900: The early construction period represents older homes built before the 20th century.

1901-1950: This period includes homes from the early to mid-20th century, which may have different architectural styles and technologies.

1951-2000: represents homes built in the latter half of the 20th century, which might reflect more modern construction methods.

2001-2010: Homes built in the 21st century.

Data Preparation

- The response variable was transformed using the square root because it is skewed to the right.
- The continuous variables were all scaled to have a mean of 0 and a standard deviation of 1, making them comparable.
- The ordinal variables were used numerically to capture how each level affects the sale price.
- The categorical variables were coded using 1-hot encoding.

The data was split into 80% training set and 20% test set

Models

Five models were built

- **Linear regression model with flat prior**
This allows the model to take any parameter without much restrictions based on the data. It does not incorporate prior beliefs on the parameters.
- **A linear regression model with weakly informative priors**
Allowing for some flexibility but ensuring that the model doesn't overfit the data but still incorporates some regularization. Parameters can deviate from prior beliefs based on the data but not extreme values.
- **Linear regression with ridge regularization (L2 penalty)**
To prevent overfitting, especially with a large number of predictors. It also penalizes extreme values.

- **Hierarchical model using year sold as a group effect**

It assumes that there is some variation in the sales price between different years of sale, which is modelled as a random effect. This helps to capture the effect of economic conditions on the sale price from the year 2006-2010.

- **Hierarchical model using year built as group effect**

It assumes that there is some variation in the sales price between the years of building, which is modelled as a random effect. This helps to capture the effect of age, architectural design structure etc on the sale price.

Model comparison

The model's performance on test data was compared and the model with the lowest RMSE was selected.